

Qualitative Analysis

If LLM P@10 > Baseline P@10 save as LLM helped

If LLM P@10 < Baseline P@10 save as LLM hurt

Results Summary

Case	Query	Model	Expected Topic	Baseline P@10	LLM P@10	Result
1	artificial intelligence in industry	BM25	Artificial Intelligence	1.0	0.8	LLM hurt
2	statistical modeling data preprocessing	BM25	Data Science	0.6	0.8	LLM helped
3	cybersecurity encryption authentication	BM25	Cybersecurity / Security	0.6	0.8	LLM helped
4	internet of things	BM25	Internet of Things	0.9	0.5	LLM hurt

Case Notes

Case	Qualitative Explanation
1	The LLM hurt performance because the original query was already clear and matched the Artificial Intelligence topic well. The LLM rewrite and expansion made the query more general, reducing Precision@10 from 1.0 to 0.8.
2	The LLM helped performance because the augmented query retrieved more documents from the Data Science topic. Precision@10 increased from 0.6 to 0.8.
3	The LLM helped performance because it rewrote the query into a more focused security-related form and expanded it with encryption protocol terms. Precision@10 increased from 0.6 to 0.8.
4	The LLM hurt performance because it rewrote "internet of things" incorrectly as a general internet/location query and expanded it with "list of places on the internet." This changed the meaning away from IoT and reduced Precision@10 from 0.9 to 0.5.

Conclusion

The qualitative analysis includes two cases where the LLM helped retrieval performance and two cases where the LLM hurt performance. The LLM helped when rewriting or expansion added useful terms that matched the expected topic. It hurt when the original query was already clear or when the LLM changed the meaning of the query and introduced irrelevant terms.